CollegeDB Semester Project Report

1. Project Overview

This project involves designing and implementing a College Database (CollegeDB) using MySQL. It includes database creation, table design, data insertion, SQL queries, views, procedures, and triggers.

2. Database Design

Seven tables were created:

- Students (student_id, first_name, last_name, email, phone, enrollment_date)
- Courses (course_id, course_name, credits)
- Teachers (teacher_id, first_name, last_name, email, dept_id)
- Departments (dept_id, dept_name)
- Enrollments (enroll_id, student_id, course_id, grade)
- CourseAssignments (assignment_id, teacher_id, course_id)
- StudentBackup (backup of Students table)

3. SQL Features Implemented

Implemented Features:

- DDL: CREATE, ALTER tables
- DML: INSERT, UPDATE, DELETE
- Queries: SELECT, WHERE, LIKE, BETWEEN, IN, ORDER BY
- Aggregates: COUNT, MIN, MAX, AVG, SUM
- GROUP BY and HAVING
- JOINS: INNER, LEFT, RIGHT, CROSS
- Set Operations: UNION, INTERSECT (via subquery), MINUS (via LEFT JOIN)
- Subqueries in WHERE and FROM
- Views
- Stored Procedures, Functions, Triggers, Cursor, Dynamic SQL

4. ER Diagram

The following page contains the ER Diagram representing all tables and relationships.

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